MAT 221 - CALCULUS I
Lesson 4 - Integration By Parts

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Integration by parts: Indefinite integral form

If u and v are functions of x, by product rule

$$\frac{d}{dx}(uv) = uv' + u'v.$$

Integrating both sides with respect to x, we have

Theorem 1

Integration by parts: Indefinite integral form

$$\int u\,dv = uv - \int v\,du \tag{1}$$

Theorem 2

Integration by parts: Definite integral form

$$\int_a^b u\,dv = uv|_a^b - \int_a^b v\,du = u(b)v(b) - u(a)v(a) - \int_a^b v\,du \tag{2}$$

Example 1

Compute

(a) $\int xe^x\,dx$ (b) $\int x^2 \ln x\,dx$ (c) $\int_0^1 x\tan^{-1}x\,dx$ (d) $\int_0^1 x^3 e^{x^2}\,dx$

Solutions

(a) to make use of eq.1, set $u = x$ and $dv = e^x\,dx$, then $du = dx$, $v = e^x$, and

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C$$

(b) Set $u = \ln x$ and $dv = x^2\,dx$, then $du = \frac{1}{x}dx$, $v = \frac{x^3}{3}$ and

$$\int x^2 \ln x\,dx = \frac{1}{3}x^3 \ln x - \frac{1}{3}\int x^2\,dx = \frac{1}{3}x^3 \ln x - \frac{1}{9}x^3 + C$$

(c) Set $u = \tan^{-1}x$ and $dv = x\,dx$, then $du = \frac{1}{1+x^2}dx$, $v = \frac{x^2}{2}$ and

$$\int_0^1 x\tan^{-1}x\,dx = \frac{1}{2}x^2\tan^{-1}x\Big|_0^1 - \frac{1}{2}\int_0^1 \frac{x^2}{1+x^2}dx$$
$$= \frac{\pi}{8} - \frac{1}{2}[x - \tan^{-1}x]_0^1 = \frac{\pi - 2}{4}$$

(d) in class

Repeated Use of Integration By Parts

Example 2

Evaluate

(a) $\int x^2 \cos 2x \, dx$ (b) $\int_1^2 \ln^2 x \, dx$ (c) $\int e^{3x} \sin 2x \, dx$

Solution

(a) Set $u = x^2$ and $dv = \cos 2x \, dx$, then $du = 2x \, dx$, $v = \frac{1}{2} \sin 2x$, and

$$\int x^2 \cos 2x \, dx = \frac{1}{2} x^2 \sin 2x - \int x \sin 2x \, dx$$

Re-applying integration by parts, we set $u = x$ and $dv = \sin 2x \, dx$, then $du = dx$, $v = -\frac{1}{2} \cos 2x$ and

$$\begin{aligned} \int x^2 \cos 2x \, dx &= \frac{1}{2} x^2 \sin 2x - \int x \sin 2x \, dx \\ &= \frac{1}{2} x^2 \sin 2x - \left[-\frac{1}{2} x \cos 2x + \frac{1}{2} \int \cos 2x \, dx \right] \\ &= \frac{1}{2} x^2 \sin 2x + \frac{1}{2} x \cos 2x - \frac{1}{4} \sin 2x + C \end{aligned}$$

(b)-(c) solutions in class